AQUABOTS Power System — Master Reference & Changelog

Team 26061 | ENGR 498B Spring 2026 Owner: John (Software Integration Lead / Power System Design) **Last Updated:** 2026-03-19 (Rev WG-1)

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1. CURRENT CONFIGURATION (Rev WG-1)

1.1 Dual-Config Concept

The wiring harness is designed ONCE for full production specs. Two battery configurations share the same harness:

Parameter	Production	Demo
Battery	LiTime 24V 100–200Ah LiFePO4	Ovonic 6S LiPo 22.2V / 6Ah
Voltage	25.6V nominal (8S LiFePO4)	22.2V nominal (6S LiPo)
Connection	M8 bolt posts → 4 AWG ring terminals	XT90 → adapter pigtail → harness
Main Fuse	MRBF **100A**	MRBF **40A** (swap element only)
Motor Fuse	MIDI **60A**	MIDI **60A** (same)
Trunk Wire	4 AWG Red/Blk (permanent harness)	10–12 AWG pigtail (adapter only)
Endurance	5+ hours (battery swaps)	~51 min (133.2 Wh / 155W)
Adapter Pigtail	Not needed	XT90 ♀ → 10–12 AWG → ring terminal (6–8″)

To switch configs: Swap MRBF fuse element + connect/disconnect adapter pigtail. No rewiring.

Demo constraint: MRBF 40A limits max system draw — motor cannot sustain 54A full throttle. This is correct/safe behavior for a 6Ah LiPo through 12 AWG leads.

1.2 Main Trunk

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Battery → MRBF Fuse (100A prod / 40A demo) → Kill Switch (Manual) → (+) Busbar
Battery (−) → (−) Busbar
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#	From	To	Gauge	~Length	Color	Fuse	Connector	Ring
P1	Batt (+)	MRBF Holder	4 AWG	6″	RED	MRBF 100A	Ring→M8	3/8″
P2	MRBF Out	Kill Switch	4 AWG	6″	RED	—	Ring→lug	3/8″
P3	Kill Switch	(+) Busbar	4 AWG	6″	RED	—	Ring→stud	1/4″
P4	Batt (–)	(–) Busbar	4 AWG	12″	BLK	—	Ring→M8	1/4″

Demo adapter pigtail (removable):

#	From	To	Gauge	~Length	Color	Fuse	Connector	Ring
D1	XT90 ♀ (+)	MRBF input stud	10–12 AWG	6–8″	RED	MRBF 40A*	XT90 ♀ →ring	3/8″
D2	XT90 ♀ (–)	(–) Busbar	10–12 AWG	6–8″	BLK	—	XT90 ♀ →ring	1/4″

*Swap MRBF element only.

1.3 Branch Circuits (All Branches — Same for Both Configs)

#	Branch	From	To	Gauge (+)	Gauge (–)	Fuse	Max Load	Cruise
P5/P6	ESC→Motor	Busbar	ESC	**8 AWG**	**8 AWG**	**MIDI 60A**	54A	60W
P7/P8	Starlink (direct)	Busbar	Starlink Mini	**12 AWG**	**12 AWG**	ATO 7.5A SB	4.7A	70W
P9/P10	5V Pwr Mod→Pixhawk	Busbar	5V Pwr Module	**12 AWG**	**12 AWG**	ATO 2A SB	~0.5A	~2W

P11/P12	12V Buck→Jetson+Fan	Busbar	12V Buck	**12 AWG**	**12 AWG**	ATO 3A SB	1.26A	15W
P13/P14	5V Buck→ESP32	Busbar	5V Buck	**12 AWG**	**12 AWG**	ATO 1A SB	0.23A	5W
P15/P16	BEC→Servo	Busbar	BEC (Castle)	**12 AWG**	**12 AWG**	ATO 2A SB	1.0A	5W

Starlink note: NO converter in path. Starlink Mini accepts 12–48V DC directly. Both 22.2V (demo) and 25.6V (prod) are in range.

1.4 Load-Side Connections (Converter → Device)

#	From	To	Gauge	~Length	Connector
P17	ESC (3-phase)	Motor (Apisqueen 5080)	8 AWG	10″	Bullet 4mm
P18	5V Power Module	Pixhawk 4X FC	22 AWG	4″	DF13 6-pin
P19	12V Buck	Jetson Orin Nano	18 AWG	6″	DC barrel
P20	12V Buck	Cooling Fan	22 AWG	4″	JST 2-pin
P21	5V Buck	ESP32-S3	22 AWG	4″	USB-C / pin hdr
P22	BEC (Castle)	Rudder Servo	22 AWG	8″	Servo 3-pin
P23	ESP32 3.3V reg	Sensors (all)	26 AWG	varies	DuPont / JST
P24	Jetson 3.3V reg	Camera (IMX219)	Ribbon	3″	15-pin FPC

1.5 Fuse Coordination Verification

Every branch passes: **Max Load < Fuse Rating < Wire Ampacity**

Branch	Max Load	Fuse	Wire	Wire Ampacity	✅
Trunk (prod)	61.3A	100A	4 AWG	135A	PASS

Trunk (demo)	40A (governed)	40A	10–12 AWG	41–55A	PASS
ESC/Motor	54A	60A	8 AWG	73A	PASS
Starlink	4.7A	7.5A	12 AWG	41A	PASS
Pixhawk	~0.5A	2A	12 AWG	41A	PASS
Jetson+Fan	1.26A	3A	12 AWG	41A	PASS
ESP32	0.23A	1A	12 AWG	41A	PASS
BEC/Servo	1.0A	2A	12 AWG	41A	PASS

1.6 Power Budget

Subsystem	Voltage	Max Power	Cruise Power	Max Current
Propulsion (Motor)	22–26V	1200W	60W	54.0A
Comms (Starlink)	22–26V direct	105W	70W	4.7A
Compute (Jetson+Fan)	12V	28W	15W	1.3A
Control (ESP32+Servo)	5V / 7.4V	22W	5W	1.0A
Payload (Sensors)	3.3V	6W	5W	0.3A
TOTAL	—	**~1361W**	**~155W**	**~61.3A**

- Demo endurance: 133.2 Wh / 155W ≈ **51 min**
- Production endurance target: **5+ hours** (battery swaps)

1.7 Signal & Data Interconnects

#	From	To	Gauge	Protocol	Signal	Connector
S1	ESP32 UART1	Pixhawk (IF-1)	22 AWG	MAVLink UART	3.3V TTL	JST-SH 4-pin
S2	ESP32 UART2	Jetson (IF-3)	22 AWG	UART 115200	3.3V TTL	JST-SH 4-pin
S3	ESP32 GPIO	ESC Signal	22 AWG	PWM 50Hz	3.3V logic	Servo 3-pin
					LVDS	

S4	Jetson CSI	Camera IMX219	Ribbon	MIPI CSI-2	diff	15-pin FPC
S5	Jetson Ethernet	Starlink Mini	Cat5e	100BASE-T	Ethernet	RJ-45
S6	ESP32 WiFi	Other Drones	—	ESP-NOW	2.4 GHz RF	Internal ant.
S7	ESP32 I2C	MPU6050 IMU	26 AWG	I2C	3.3V	DuPont 4-pin
S8	ESP32 I2C	Atlas pH Circuit	26 AWG	I2C / UART	3.3V	DuPont + BNC
S9	ESP32 1-Wire	DS18B20 Temp	26 AWG	1-Wire	3.3V	Waterproof
S10	ESP32 Analog	TDS Conductivity	26 AWG	Analog	0–3.3V	DuPont 2-pin
S11	ESP32 UART	GPS Module	26 AWG	UART 9600	3.3V TTL	DuPont 4-pin
S12	ESP32 GPIO	DFRobot Ultrasonic	26 AWG	Digital IO	3.3V / 5V	DuPont 4-pin

1.8 Fuse Summary (Quick Reference)

Holder	Production	Demo	Type	Protects
MRBF	**100A**	**40A** (swap)	Bolt-down	Main trunk
MIDI	**60A**	**60A**	Bolt-down	ESC / Motor
ST Blade ATO	7.5A SB	7.5A SB	Blade	Starlink Mini (direct)
ST Blade ATO	2A SB	2A SB	Blade	5V Pwr Mod → Pixhawk
ST Blade ATO	3A SB	3A SB	Blade	12V Buck → Jetson + Fan
ST Blade ATO	1A SB	1A SB	Blade	5V Buck → ESP32
ST Blade ATO	2A SB	2A SB	Blade	BEC → Rudder Servo

SB = Slow-Blow

1.9 EMI Filtering (Rev ISR-4b)

All EMI filtering components, placement locations, and soldering instructions are documented in **AQUABOTS_EMI_FILTERING_BOM.pdf (Rev ISR-4b)**. Summary below:

Busbar-Level Bypass

Part	Value	Placement	Notes
C1	1000uF / 35V+ Electrolytic	Across (+) and (–) busbar studs, <1″ from stud centers	Absorbs low-freq transients (motor startup, ESC bursts)
C2	100nF / 50V+ Ceramic (ON HAND)	In PARALLEL with C1, same studs	Absorbs high-freq hash (>1MHz). Both caps required. Verify 50V+ rating for busbar-side caps.

Per-Branch Filtering

Branch	Input-Side Ferrite	Output-Side Cap(s)	Notes
ESC/Motor (P5)	F1: Hondark MnZn toroid 19mm ID (38x19x13mm) on 8 AWG (+), <2″ from ESC	C3: 470uF/35V+ **LOW-ESR** + C2: 100nF parallel at ESC input	Biggest noise source. Toroid = single pass. Low-ESR critical HERE ONLY.
5V Buck→ESP32 (P13/P21)	F2: Raogoodcx 13mm clip-on on 12 AWG (+) input	F4: 13mm clip-on on 22AWG output pair (both wires, common-mode); C4: 10uF/10V + C2: 100nF parallel at ESP32 VIN	Most sensitive branch. Both input AND output filtering.
12V Buck→Jetson (P11/P19)	F2: Raogoodcx 13mm clip-on on 12 AWG (+) input	C5: 100uF/50V (ON HAND) + C2: 100nF parallel at Jetson DC barrel	Jetson has onboard regulation but needs clean input.
Starlink (P7)	F3: Raogoodcx 13mm clip-on on 12 AWG (+)	C7: 100uF/50V (ON HAND) + C2: 100nF parallel at Starlink DC plug	Direct from busbar, no converter — needs busbar noise filtering.
BEC→Servo (P15/P22)	— (none needed)	C6: 100uF/50V (ON HAND) + C2: 100nF parallel at servo connector	Servo is inductive noise source; cap at output absorbs

			it.
Pixhawk (P9)	— (none needed)	— (Pixhawk power module has internal filtering)	Lowest-current branch, well-isolated by 5V power module.

Sensor Signal Filtering

Signal	Filter	Placement
S10 (TDS Analog)	C2: 100nF ceramic (ON HAND), pin-to-GND	<0.25″ from ESP32 ADC pin. **CRITICAL** — analog most vulnerable to ESC noise.
S7 (IMU I2C)	F5: BOJACK axial bead ~70Ω@100MHz inline on SDA and SCL	<1″ from ESP32 I2C pins. Cut wire, solder bead in series. 1 bead per line sufficient.
S8 (pH I2C)	F5: BOJACK axial bead ~70Ω@100MHz inline on SDA and SCL	<1″ from ESP32 I2C pins. Same type as S7. 4 beads total for S7+S8.
S9 (1-Wire)	C2: 100nF ceramic (ON HAND), pin-to-GND (optional)	<0.5″ from ESP32 pin. Lower priority than S10/S7/S8.
S11 (GPS UART)	C2: 100nF ceramic (ON HAND), pin-to-GND (optional)	<0.5″ from ESP32 pin. Lower priority.

Note: 600Ω@100MHz through-hole ferrite beads do not exist as a consumer product. ~70Ω generic axial beads are adequate for short I2C runs combined with 100nF bypass caps.

Copper Tape Cable Shielding

Cable	Priority	Ground Drain
P21: 5V Buck output → ESP32 (22 AWG)	HIGH	Drain to 5V buck GND terminal (source end)
S10: TDS Analog (26 AWG, 12″ run)	HIGH	Drain to ESP32 GND (source end)
S7-S9, S11: Sensor bundle (26 AWG)	MEDIUM	Drain to ESP32 GND (single drain for bundle)
S1: UART to Pixhawk (22 AWG)	MEDIUM	Drain to ESP32 GND

S2: UART to Jetson (22 AWG)	MEDIUM	Drain to ESP32 GND
P5/P6: ESC-Motor 3-phase (8 AWG)	SKIP	Noise source — ferrite + caps handle it
P1-P4: 4 AWG trunk	SKIP	Short, fat, low-impedance — not worth shielding

Shielding rules: Ground copper tape at ONE end only (source end). Ungrounded = antenna. Both ends grounded = ground loop. Wrapping order: copper tape (innermost) → automotive self-adhesion tape (insulation) → split loom wrap (mechanical protection). Leave 0.5″ bare wire at connectors to avoid shorts.

Key rules: Parallel caps = electrolytic + ceramic on same solder points. Ferrite clip-ons go on (+) wire only (except F4 output which takes both wires for common-mode). All leads as short as possible. Voltage rating is a maximum, not a target — 50V caps on 5V rails are fine. Secure everything with zip ties for marine vibration.

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2. CHANGELOG

Rev WG-1 (2026-03-19) — 12 AWG Full Standardization + Wiring Guide + CAT6 Bow Sensor Cable

Created comprehensive wiring integration guide (AQUABOTS_WIRING_GUIDE.typ / .pdf) as the authoritative build document.

1. **All non-motor busbar branches standardized to 12 AWG**
   - P7/P8 (Starlink): 16 AWG → **12 AWG**
   - P9/P10 (Pixhawk 5V Pwr Module): 18 AWG → **12 AWG**
   - P11/P12 (12V Buck→Jetson): 18 AWG → **12 AWG**
   - P13/P14 (5V Buck→ESP32) and P15/P16 (BEC→Servo) were already 12 AWG since Rev ISR-3.
   - Rationale: Marine durability, inventory simplification (one spool for all branches).
   - Fuse coordination still passes for all branches (12 AWG rated 41A, all fuses ≤7.5A).
2. **Shielded CAT6 cable adopted for bow sensor bundle**
   - Pre-made shielded CAT6 patch cable (~$6) replaces individual 26 AWG copper-tape-wrapped sensor wires for the long bow run (18–24″).
   - Uses all 4 twisted pairs: Pair 1 = pH I2C SDA/GND, Pair 2 = pH I2C SCL/VCC, Pair 3 = Temp 1-Wire/GND, Pair 4 = TDS Analog/GND.

- CAT6 foil shield grounded at ESP32 end only (source end). No copper tape needed on this bundle.
- Rationale: Superior EMI rejection for analog TDS and I2C signals over 18–24″ through noisy hull. Per-pair twisting + overall foil shield outperforms copper tape on individual wires.

3. **Section 1.3 and 1.5 tables corrected** to reflect 12 AWG standardization (were stale since Rev ISR-3 only updated P13/P14 and P15/P16).
4. **Wiring guide created as new deliverable:**
   - `AQUABOTS_WIRING_GUIDE.typ` — Typst source for local compilation
   - `AQUABOTS_WIRING_GUIDE.pdf` — Pre-compiled PDF
   - Contains: complete wire-by-wire tables with gauges/lengths/connectors, CAT6 pair allocation, EMI filter placement, assembly sequence, wire cut list, routing channels.
   - This guide supersedes the changelog tables for build purposes.

Rev ISR-4b (2026-03-04) — EMI Filtering Addition + Refinements

Added EMI filtering across all power branches, critical signal lines, and copper tape shielding:

1. **Busbar bypass caps: 1000uF/35V+ electrolytic + 100nF ceramic in parallel**
   - Placed across (+) and (–) busbar studs to absorb low- and high-frequency transients.
   - This is the single most important EMI mitigation — catches ESC-generated noise at the source bus.
2. **ESC/Motor branch: Ferrite toroid + LOW-ESR parallel caps at ESC input**
   - Hondark MnZn ferrite toroid (38x19x13mm, 19mm ID) on 8 AWG (+) lead, single pass, within 2″ of ESC.
   - 470uF/35V+ **LOW-ESR** electrolytic + 100nF ceramic in parallel at ESC power input terminals.
   - Low-ESR is critical at this position ONLY — high-current transients from ESC switching.
3. **5V Buck→ESP32 branch: Dual-stage filtering (input + output)**
   - Input: Raogoodcx 13mm clip-on ferrite on 12 AWG (+) lead, within 2″ of buck input.
   - Output: Clip-on ferrite on 22 AWG pair (both wires, common-mode), within 2″ of ESP32.
   - 10uF/10V+ electrolytic + 100nF ceramic in parallel at ESP32 VIN pins.

- Rationale: ESP32 2.4GHz RF and analog ADC inputs are the most EMI-sensitive loads.

4. **12V Buck→Jetson branch: Input ferrite + output caps**
   - Raogoodcx 13mm clip-on ferrite on 12 AWG (+) input lead.
   - 100uF/50V electrolytic (on hand) + 100nF ceramic in parallel at Jetson DC barrel.
5. **Starlink branch: Input ferrite + output caps**
   - Raogoodcx 13mm clip-on ferrite on 12 AWG (+) lead after ATO fuse.
   - 100uF/50V electrolytic (on hand) + 100nF ceramic at Starlink DC input connector.
6. **BEC→Servo branch: Output caps only**
   - 100uF/50V electrolytic (on hand) + 100nF ceramic at servo connector output.
7. **Sensor signal filtering**
   - 100nF ceramic cap pin-to-GND on TDS analog input (S10) — CRITICAL for ADC accuracy.
   - BOJACK ~70Ω@100MHz axial ferrite beads inline on I2C SDA/SCL for IMU (S7) and pH (S8) — 4 beads total.
   - Note: 600Ω@100MHz through-hole beads don't exist as consumer product. ~70Ω adequate for short I2C runs.
   - Optional 100nF caps on 1-Wire (S9) and GPS UART (S11) signal pins.
8. **Copper tape cable shielding (new in 4b)**
   - HIGH priority: P21 (5V buck → ESP32) and S10 (TDS analog) — copper tape wrapped, grounded at source end.
   - MEDIUM priority: Sensor bundle (S7-S9, S11), UART lines (S1, S2) — copper tape wrapped, grounded at ESP32 GND.
   - SKIP: ESC-motor 3-phase (noise source, not victim) and 4 AWG trunk (short, low-impedance).
   - Critical rule: Ground shield at ONE end only (source end). Ungrounded = antenna. Both ends = ground loop.
   - Wrapping order: copper tape → automotive self-adhesion tape → split loom wrap.

Parts on hand (no order needed): C5, C6, C7 (100uF/50V electrolytics), C2 (100nF ceramics — verify 50V+ for busbar-side), automotive tape, loom wrap. **Parts to order:** C1, C3 (low-ESR), C4, F1 (Hondark toroid), F2-F4 (Raogoodcx 13mm clip-ons 15-pack), F5 (BOJACK axial beads 100-pack), copper foil tape. **Estimated order cost:** ~$50 (includes generous spare counts on

ferrites/beads). **Documentation:** Full BOM with placement instructions in AQUABOTS_EMI_FILTERING_BOM.pdf (Rev ISR-4b). **Wiring diagram updated:** AQUABOTS_WIRING_DIAGRAM.html now shows all EMI components (Rev ISR-4).

Rev ISR-3 (2026-03-04) — Fuse Coordination Audit + Wire Corrections

Critical fixes:

1. **Motor branch wire gauge corrected: 12 AWG → 8 AWG**
    - Root cause: ISR slide 27 showed 8 AWG Red / 10 AWG Blk to ESC. Project summary incorrectly stated "12 AWG (upgraded from 8/10 AWG)" — the ISR judge's "upgrade to 12 AWG" was for non-motor branches (from 14 AWG), NOT the motor branch.
    - 12 AWG (41A rated) cannot safely carry 54A motor max draw. 8 AWG (73A) provides proper headroom.
    - Both (+) and (–) standardized to 8 AWG for motor branch.
2. **Motor fuse downsized: MIDI 125A → MIDI 60A**
    - 125A fuse on 12 AWG wire was a critical fuse coordination failure (fuse > wire ampacity).
    - 60A fuse properly coordinates: 54A load < 60A fuse < 73A wire (8 AWG).
    - 60A slow-blow MIDI handles motor startup inrush without nuisance tripping.
3. **Main trunk fuse downsized: MRBF 150A → MRBF 100A (production)**
    - 150A exceeded 4 AWG wire ampacity (135A). Fuse would not protect wire in a dead short.
    - 100A properly coordinates: 61.3A load < 100A fuse < 135A wire (4 AWG).
4. **Demo trunk fuse adjusted: MRBF 50A → MRBF 40A**
    - 50A fuse on 10–12 AWG pigtail (41–55A) was marginal. 40A ensures wire protection.
    - 40A also acts as safety governor — prevents sustained full-throttle on 6Ah LiPo.

Minor improvements:

5. **Pixhawk branch fuse: ATO 1A → ATO 2A SB**
    - 1A was tight for Pixhawk power module inrush. 2A provides margin.
6. **Jetson branch fuse: ATO 7.5A → ATO 3A SB**
    - 7.5A was 6× the 1.26A max load. 3A provides tighter fault protection.
7. **5V Buck→ESP32 wire upgraded: 18 AWG → 12 AWG**

    - Per John’s request for production marine durability and wire inventory standardization.
8. **BEC→Servo wire upgraded: 18 AWG → 12 AWG**
    - Same rationale as above.
9. **Starlink connection clarified: Direct from busbar (no converter)**
    - Starlink Mini accepts 12–48V DC. Both 22.2V (demo) and 25.6V (prod) are in range.
    - Path: (+) Busbar → ATO 7.5A SB → Starlink Mini. No voltage conversion.

Rev ISR-2 (2026-03-04) — Dual Configuration Design

- Added production (LiTime LiFePO4) / demo (Ovonic LiPo) dual-config concept.
- Designed XT90 adapter pigtail for demo battery connection.
- Established “build once for production, adapt for demo” principle.

Rev ISR (2026-02-xx) — ISR Presentation

- Main fuse upgraded: 80A → 150A (MRBF holder).
- Added dedicated 125A motor fuse (MIDI block).
- Added proper fuse blocks: MRBF + MIDI + ST Blade ATO.
- Added thermal pads for heat management.
- All parts ordered ($417 total).

Rev CDR (2025-12-xx) — Major Simplification

- Switched from 24V LiFePO4 to 6S LiPo (22.2V).
- Eliminated 48V boost converter and PoE injector.
- Eliminated BMS.
- Starlink Mini powered directly from battery rail.

Rev PDR (2025-10-xx) — Original Design

- 24V LiFePO4 with BMS.
- 48V boost converter + PoE for Starlink.
- 19V buck converter.
- Complex power chain (deprecated).

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3. PARTS LIST — WHAT TO ORDER / SWAP

Fuse Elements to Order (replacing previously ordered items)

Part	Replaces	Qty	Est. Price
MRBF 100A fuse element	MRBF 150A (ordered)	1	~$8
MRBF 40A fuse element	(new for demo)	1	~$8
MIDI 60A fuse element	MIDI 125A (ordered)	1	~$6
ATO 2A slow-blow	ATO 1A (Pixhawk branch)	1	~$2
ATO 3A slow-blow	ATO 7.5A (Jetson branch)	1	~$2

Wire to Order

Item	Qty	Notes
8 AWG Red silicone wire	2 ft	ESC (+) branch
8 AWG Black silicone wire	2 ft	ESC (–) branch
12 AWG Red (if not already stocked)	3 ft	5V Buck + BEC branches
12 AWG Black (if not already stocked)	3 ft	5V Buck + BEC branches

Ring Terminals to Order

Size	For	Qty
8 AWG ring, 1/4″ stud	ESC branch to busbar	2
8 AWG ring, MIDI blade	ESC branch to MIDI holder	1

Existing Parts (No Change)

- MRBF fuse holder — keep (just swap element)
- MIDI fuse block — keep (just swap element)
- ST Blade ATO fuse block — keep
- Bus bars (+/–) — keep

- Kill switch — keep
- Castle BEC — keep
- All buck converters — keep
- All connectors — keep

EMI Filtering Parts (Rev ISR-4b)

To Order:

Part	Value / Spec	Product	Qty	Est. Price
C1	1000uF / 35V+ radial electrolytic	Any standard	1	~$2
C3	470uF / 35V+ **LOW-ESR** radial electrolytic	Low-ESR specifically — only cap where ESR matters	1	~$3
C4	10uF / 10V+ radial electrolytic	Any standard	1	~$0.50
F1	Ferrite toroid, 19mm ID, MnZn	Hondark 38x19x13mm (Amazon, 4-pack)	1 pack	~$13
F2-F4	13mm clip-on ferrite	Raogoodcx 13mm (Amazon, 15-pack)	1 pack	~$16
F5	Axial ferrite bead ~70Ω@100MHz	BOJACK 3.5x6x0.8mm (Amazon, 100-pack)	1 pack	~$7
SH1	Copper foil tape, 1″ wide, conductive adhesive	Amazon	1 roll	~$8

Order subtotal: ~$50 (includes generous spare counts on ferrites and beads)

On Hand (no order needed):

Part	Value	Qty Needed	Notes
C2	100nF / 50V+ ceramic disc (through-hole)	10+	Verify 50V+ rating for busbar-side. 25V+ OK for 5V/12V side.
C5, C6,	100uF / 50V radial	3	Substituting for originally-spec'd 25V/16V/35V. 50V

C7	electrolytic		rating is fine — voltage rating is a max, not a target.
SH2	Automotive electrical self-adhesion tape	1 roll	Insulation layer over copper tape.
SH3	Split loom wrap (1/4″ and 3/8″)	as needed	Outer mechanical protection layer.

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4. ASSEMBLY NOTES

Safety-First Sequence

1. **Verify all connections are de-energized** before starting.
2. Build the trunk FIRST: MRBF holder → Kill Switch → (+) Busbar.
3. Connect (–) Busbar run.
4. **Install busbar bypass caps (E1):** Solder C1 (1000uF/35V) + C2 (100nF) in parallel across (+) and (–) busbar studs. Leads <1″. Secure with zip tie.
5. Install branch fuses in ST Blade ATO block (but leave main MRBF and MIDI empty).
6. Wire all branch circuits from busbars to converters/loads.
7. **Install input-side ferrites:** F1 toroid on ESC 8 AWG (+), F2 clip-ons on 5V Buck and 12V Buck 12 AWG (+) inputs, F3 clip-on on Starlink 12 AWG (+).
8. Wire all load-side connections (converters → devices).
9. **Install output-side ferrite:** F4 clip-on on 5V Buck → ESP32 22 AWG pair.
10. **Install all output-side parallel caps:** C3+C2 at ESC, C4+C2 at ESP32, C5+C2 at Jetson, C6+C2 at servo, C7+C2 at Starlink. Observe electrolytic polarity.
11. **Install sensor signal filtering:** C2 on TDS analog pin (S10), ferrite beads inline on I2C SDA/SCL (S7, S8).
12. **Final inspection:** Check every terminal for tightness, correct polarity, no stray strands. Verify no shorts introduced by cap leads.
13. Insert MIDI 60A fuse element.
14. Insert MRBF fuse element LAST (this energizes the system).
15. Test with multimeter at each load point before powering devices.

Demo vs Production Switchover

To switch from production → demo:

1. Remove MRBF fuse element (de-energize system).
2. Disconnect 4 AWG trunk leads from battery M8 posts.
3. Connect XT90 adapter pigtail ring terminals to MRBF input stud and (–) busbar.
4. Insert MRBF 40A fuse element.
5. Plug Ovonic LiPo XT90 into adapter pigtail.

To switch from demo → production:

1. Remove MRBF fuse element (de-energize system).
2. Disconnect adapter pigtail from MRBF stud and busbar.
3. Bolt 4 AWG trunk ring terminals to LiTime M8 posts.
4. Insert MRBF 100A fuse element.

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5. KNOWN ISSUES & FUTURE WORK

- Verify Starlink Mini DC input connector type — may need custom DC pigtail from ATO fuse block output to Starlink's proprietary power connector.
- Measure actual motor inrush current during pool testing — confirm 60A MIDI doesn't nuisance-trip on startup. If it does, consider 70A MIDI.
- Measure actual Jetson boot-up inrush — confirm 3A ATO doesn't nuisance-trip.
- Perform full-load voltage drop test across busbars under 54A motor draw.
- Add waterproofing / conformal coating plan for marine environment.
- Document LiTime battery mounting and M8 terminal torque specs.
- Verify ESP-NOW RF performance after EMI filtering installation — check RSSI and packet loss vs. pre-filter baseline.
- Test TDS analog sensor accuracy with motor running at cruise — compare with/without 100nF cap on ADC pin.
- Consider adding 100nF caps to ESP32 UART lines (S1 MAVLink, S2 Jetson) if communication errors observed during motor operation.